

MONTROSE, CO, USA

WMO: 724765

Lat: 38.506N

Lon: 107.899W

Elev: 5720

StdP: 11.90

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93013

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.0	8.7	-3.5	5.6	11.4	0.5	7.0	18.8	23.4	40.8	18.8	38.8	6.9	150	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	93.4	59.3	91.2	58.8	89.3	58.3	64.0	82.0	62.6	80.4	61.6	79.6	9.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	94.5	66.7	57.4	87.2	66.7	55.4	81.0	68.2	32.1	80.6	31.1	80.2	30.2	79.4	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	19.2	16.4	DB	-3.4	98.3	5.4	3.6	-7.2	100.9	-10.4	103.0	-13.4	105.1	-17.3	107.7	(4)
(5)			WB	-4.1	66.3	5.2	3.0	-7.9	68.5	-10.9	70.2	-13.8	71.9	-17.6	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	27.9	33.6	42.5	49.1	58.4	69.0	74.3	71.4	63.3	50.6	38.6	27.8		
	DBStd	17.57	7.91	8.03	7.71	7.36	7.57	6.23	3.99	3.53	6.37	7.50	8.13	8.57		
	HDD50	2644	686	458	252	105	19	0	0	0	4	83	348	689		
	HDD65	5962	1151	878	698	477	226	34	2	3	99	447	793	1154		
	CDD50	2875	0	0	19	79	278	571	755	663	403	102	5	0		
	CDD65	716	0	0	0	0	20	155	291	201	48	1	0	0		
	CDH74	10079	0	0	2	34	561	2609	3573	2360	880	60	0	0		
	CDH80	4323	0	0	0	2	130	1191	1775	978	243	4	0	0		
(14) Wind	WSAvg	7.0	5.3	6.1	7.5	8.5	8.4	8.5	7.4	6.9	7.1	6.7	6.1	5.2		
(15) Precipitation	PrecAvg	9.8	0.5	0.4	0.7	0.8	0.9	0.6	0.9	1.2	1.2	1.1	0.8	0.6		
	PrecMax	15.0	2.0	1.0	2.2	2.8	2.5	2.2	3.0	3.2	3.3	2.6	2.8	2.0		
	PrecMin	5.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0		
	PrecStd	2.2	0.4	0.3	0.6	0.6	0.6	0.6	0.7	0.7	0.9	0.7	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	60.9	72.5	78.0	89.6	96.1	97.3	93.4	89.8	79.5	66.8	55.2		
		MCWB	39.2	42.4	46.9	49.7	56.5	59.2	60.7	59.8	59.4	53.3	45.7	40.1		
	2%	DB	47.7	55.4	67.6	73.5	83.3	92.8	94.6	91.0	85.7	75.0	62.8	49.9		
		MCWB	36.7	40.0	44.9	47.1	53.4	57.7	60.2	59.4	56.8	51.2	44.3	38.7		
	5%	DB	44.6	51.6	63.5	70.4	80.4	90.2	92.2	88.8	82.4	71.4	58.6	45.4		
		MCWB	35.3	38.4	42.9	46.1	51.8	56.3	59.7	59.3	56.0	49.9	42.6	35.8		
	10%	DB	40.4	47.1	59.3	66.4	76.8	87.7	90.0	86.0	79.8	66.5	54.1	41.0		
		MCWB	33.0	36.5	41.2	44.9	50.5	55.5	59.6	59.3	55.3	48.0	40.6	33.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	44.0	48.1	50.9	57.5	62.2	65.8	66.4	62.3	56.4	47.7	42.2		
		MCDB	48.9	56.5	68.3	72.2	83.9	87.5	85.9	84.6	78.6	70.1	63.3	52.5		
	2%	WB	38.1	41.2	45.8	48.9	55.2	60.0	64.2	64.0	60.3	53.5	45.3	39.3		
		MCDB	44.9	53.5	65.1	69.6	77.7	84.9	83.3	80.4	76.2	69.0	59.8	47.4		
	5%	WB	35.9	39.1	43.7	47.3	53.4	58.6	63.0	62.5	58.9	51.5	43.4	36.8		
		MCDB	42.7	49.4	61.1	66.8	75.0	83.4	81.6	79.0	74.3	66.6	56.8	44.3		
	10%	WB	33.7	37.2	41.9	45.8	51.7	57.1	62.0	61.5	57.4	49.8	41.3	34.1		
		MCDB	39.9	46.2	57.5	64.3	73.0	81.7	80.6	78.2	73.5	64.4	52.9	40.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	21.7	25.9	26.3	28.2	31.7	29.7	28.6	28.7	27.3	24.6	20.6		
		MCDBR	24.2	28.0	32.8	33.2	33.7	35.3	33.9	33.3	33.3	33.4	30.7	24.0		
	5% WB	MCWBR	15.1	15.3	15.7	14.7	13.5	13.4	12.1	12.2	13.0	15.2	16.4	14.7		
		MCWBR	20.7	24.4	30.0	30.2	31.3	32.0	27.4	26.4	26.7	28.4	28.5	21.7		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.247	0.279	0.303	0.316	0.314	0.361	0.356	0.309	0.281	0.252	0.243		
	taud		2.579	2.586	2.507	2.469	2.485	2.484	2.406	2.410	2.547	2.593	2.612	2.571		
	Ebn at Noon		307	318	314	310	305	304	289	288	299	300	300	298		
	Edh at Noon		22	25	31	34	34	34	37	36	29	25	22	21		
(44) All-Sky Solar Radiation	RadAvg		835	1089	1535	1888	2158	2489	2197	1931	1706	1281	921	713		
	RadStd		55	80	122	115	167	156	126	90	92	103	67	57		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.83	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page